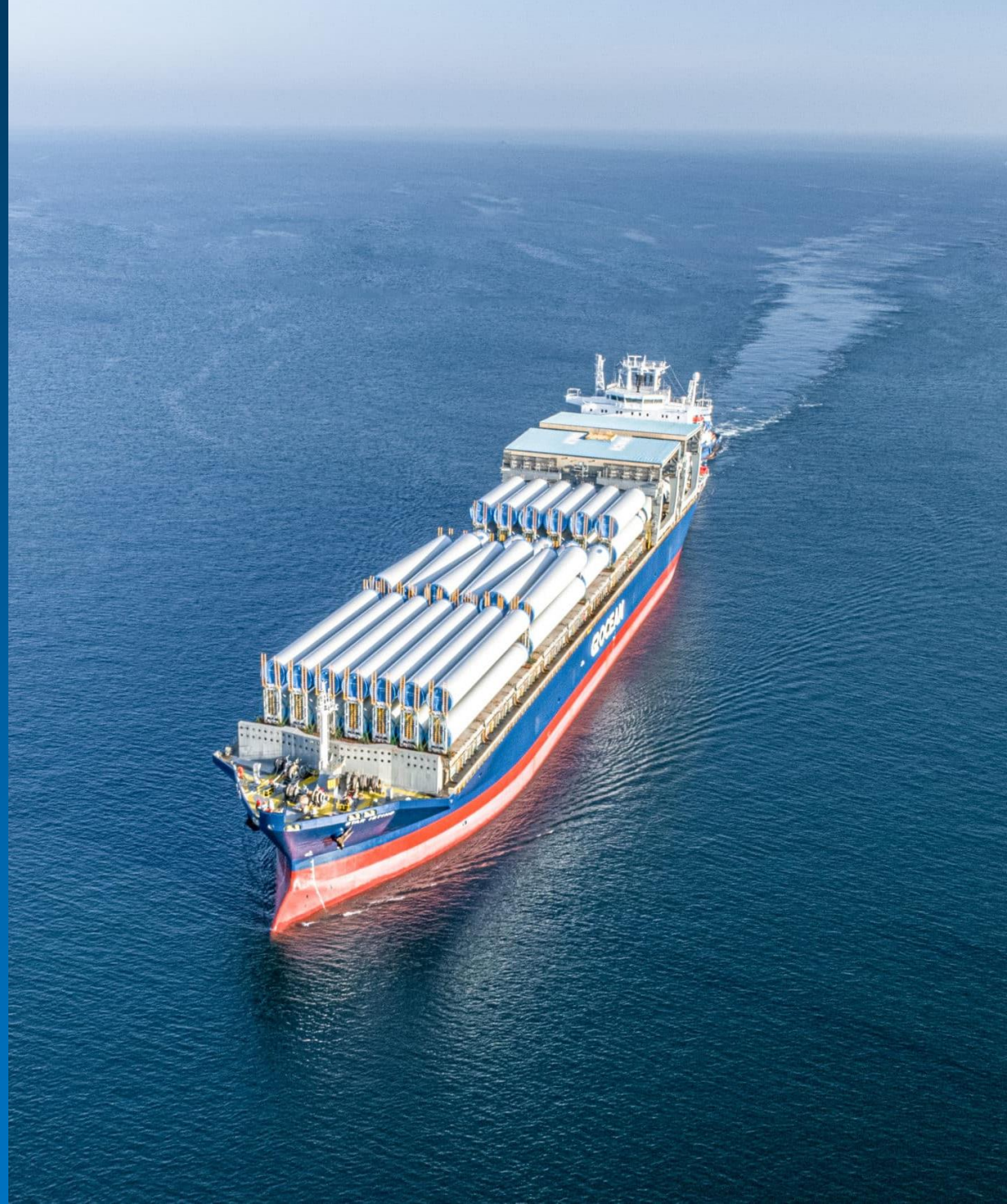
SINTEF

Operational data from ships

Real vessel data challenge





Sensor data from ships

- State-of-the-art systems: Collect, and store large amounts of sensor data
- Common data quality issues:
  - Time synchronizing mistakes
  - Unit and frequency inconsistency
  - Random missing data
  - Extended periods of missing data
  - Sensor drifting
  - Complex marine environment effects (e.g., hull fouling, weather conditions)

Picture from: https://griegmaritime.com/our-business/shipping/ship-management/



Ship performance

- **Ship performance:**
  - How energy effectively and efficiently a vessel executes its operational tasks.
    - -> How much power and fuel is being consumed?
- Why is ship performance analysis important?
- **Increased economic pressure:**
  - More unstable world situation -> increased volatility in global energy prices.
  - Fuel has become a **dominant factor in ship operational costs**
- **Increased environmental pressure:**
  - IMO have a net-zero target within 2050
  - Shipping companies face stricter regulatory requirements to reduce energy use
- **-> Improving operational efficiency** is important
  - -> Ship performance analysis is a key driver

Picture from: https://griegmaritime.com/our-business/shipping/ship-management/



Ship performance

- **Reliable ship performance analysis**
  - -> Depends on transforming **raw data into high-quality data** by using suitable data processing techniques.

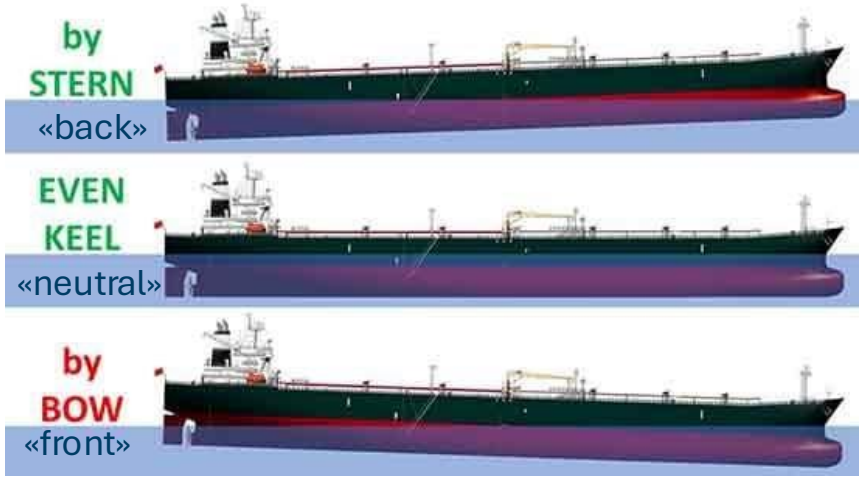
Picture from: https://griegmaritime.com/our-business/shipping/ship-management/

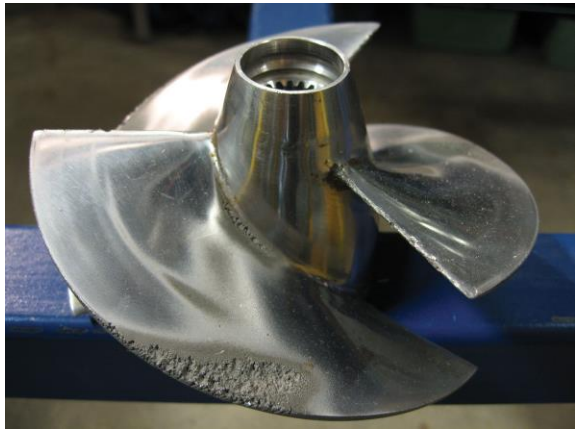
What impacts ship performance ? 1/3

Hydrodynamics:

- **Hull Roughness & Fouling:**
  - Marine growth and coating breakdown increase frictional resistance, raising required power over time.
- **Trim and Draft:**
  - The ship's draft (depth in water) and longitudinal angle (trim) change the wetted surface area and wave-making resistance
- **Propeller Efficiency:**
  - Propeller degradation, cavitation, or wear affect thrust and torque conversion relative to engine powe







What impacts ship performance ? 2/3

Environmental factors:

- **Wind Resistance:**
- **Waves:**
- **Currents**





What impacts ship performance ? 3/3

Operational modes:

- **Modes and trips:** Different operational modes require different amount of energy
  - E.g. acceleration -> transit -> deceleration
- **Loading Condition:** Cargo weight and distribution alter displacement and the ship's baseline resistance profile.

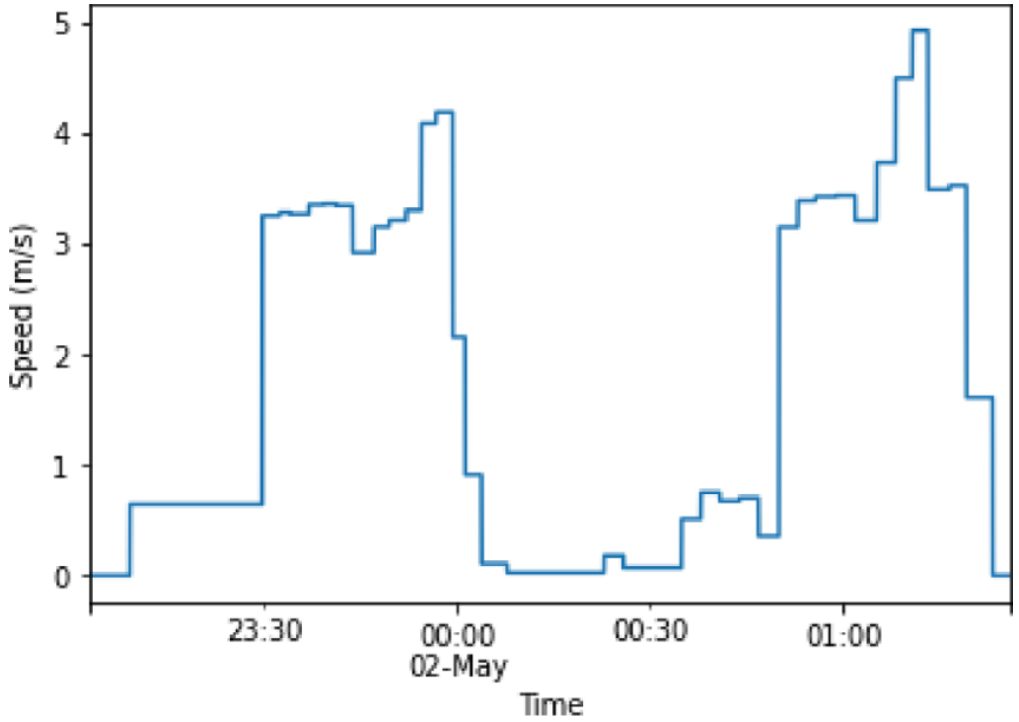


Figure from: https://www.researchgate.net/publication/371208163_Self-labelling_of_tugboat_operation_using_unsupervised_machine_learning_and_intensity_indicator



Real Vessel Data Challenge

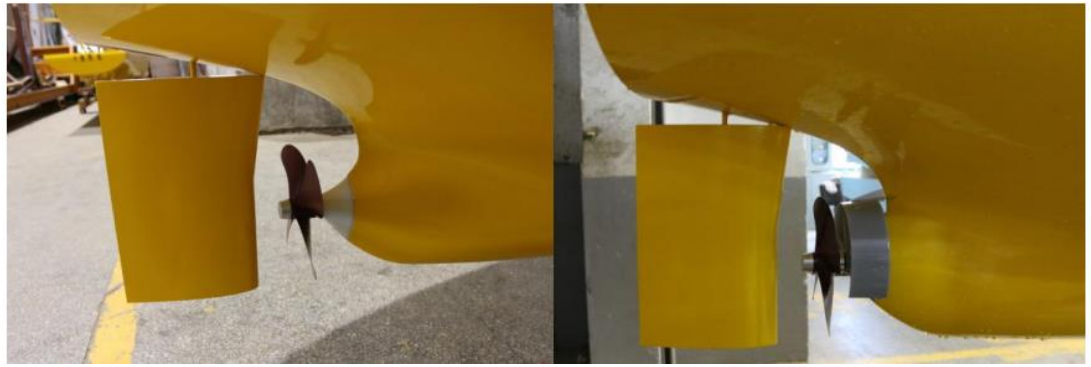
A first of a kind Workshop on Modelling, Insights and Actionable Outcome

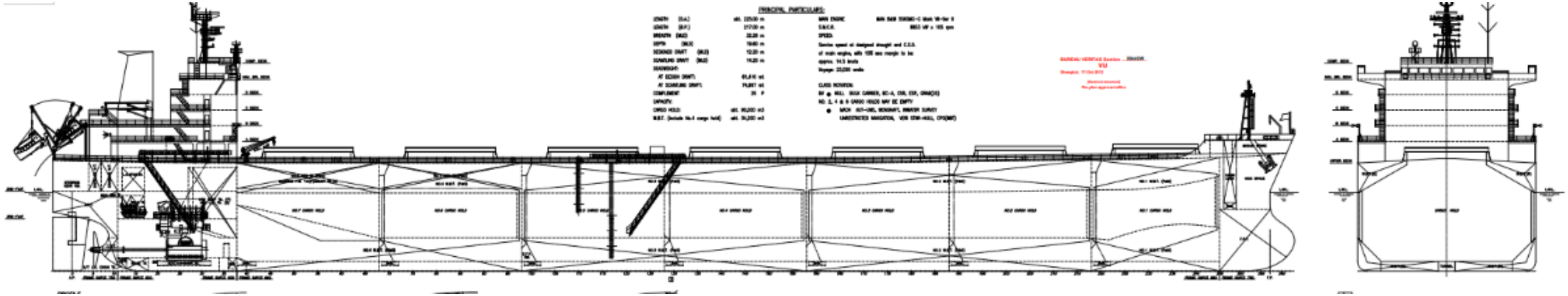
Real vessel data challenge



SHIP

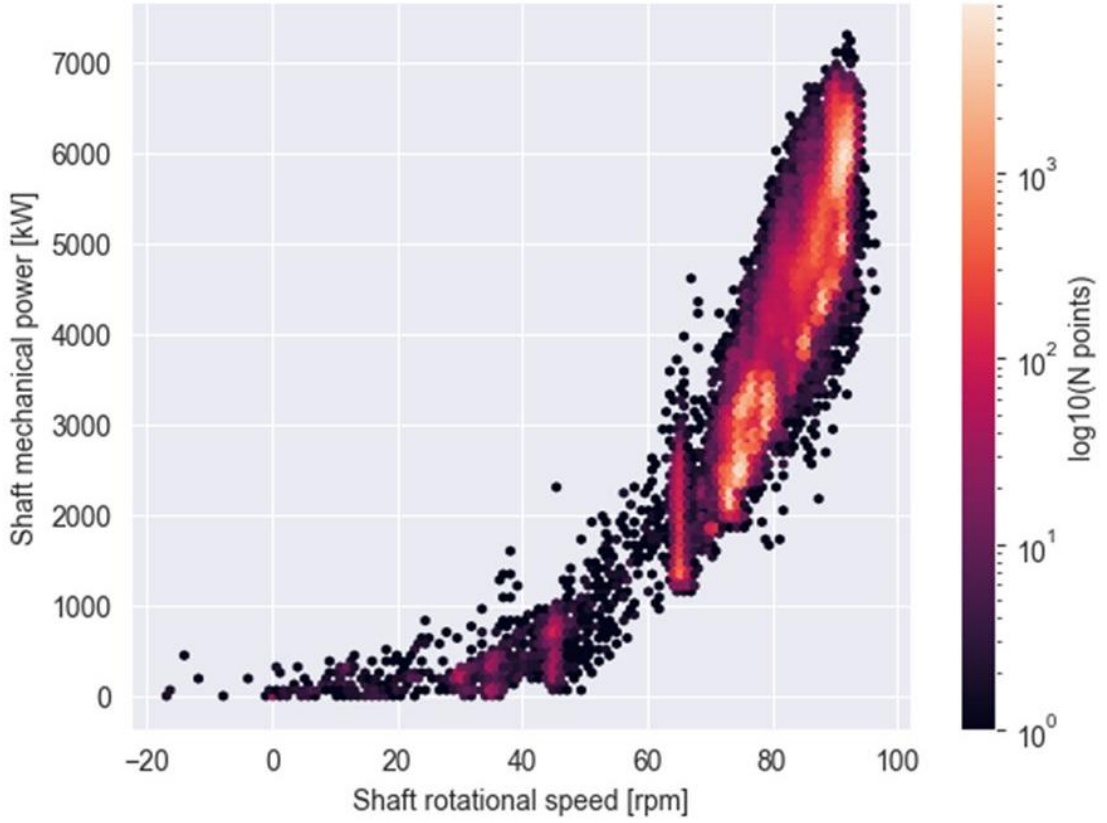
- Competition 2025: Compare methods for prediction of ship performance
- Operational data of one year from one vessel
- Vessel: Bulk carrier «Kensas»





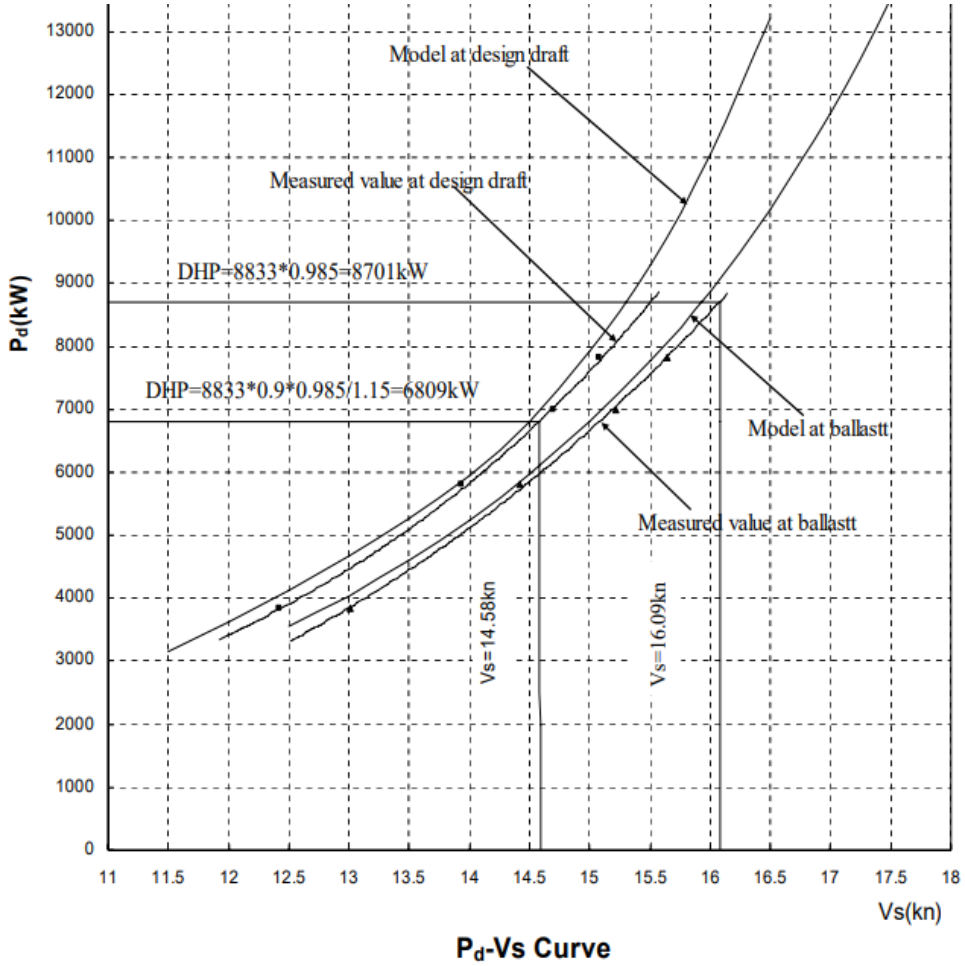
Data set

- Around 50 timeseries available
  - Weather
    - E.g.: Wind speed, wind direction, current and waves
  - Engine
    - E.g.: Power, fuel, rotation of propellers
  - Motion
    - E.g.: Speed over ground, speed through water
- 470 000 datapoints
  - One year of data
  - 1 min resolution of data
- Additionally: Cleaning events



Spesifics of the vessel

DESIGNATION	SYMBOL	Heavy Ballast
Length overall	Loa(m)	224.92
Length on waterline	L_{WL}(m)	217.91
Length between perpendicular	L_{PP}(m)	217.00
Breadth moulded on WL	B(m)	32.25
Wetted surface area	S(m^2)	8485
Displacement volume moulded	(m^3)	39024
Draft moulded on FP	T_F(m)	4.40
Draft moulded on AP	T_A(m)	9.40
Proj. area above WL	At(m^2)	836
Area of bilge keels	S_{bk}(m^2)	110
Block coefficient	CB	0.8082
LCB position of midship	LCB(lpp%)	-1.15%
Length-breadth ratio	L_{PP}/B	6.7287
Breadth-draft ratio	B/T	4.6739



P_d-Vs Curve

Sea trial curve

Data code of conduct:

- **Respect the Data Source**:
  - The dataset is provided by a real shipowner. Participants must not attempt to reverse-engineer or disclose the identity of the donor vessel.
  - **Open Results with Proper Attribution**: The source of the data is properly cited as part of the *Real Vessel Data Challenge*.
- **Controlled Sharing**: The dataset may be shared if one agree with all terms of the code.

What type of questions do we want answered?

- Data processing: Removal of outliers and sensor drifting:
  - Do you see some sensor faults that would impact the energy analysis?
    - E.g. Torque data is mis calibrated in parts of the dataset.
- **Energy analysis:** What is the ship performance during the trip?
  - Do you see any patterns in **good and bad trips** with regard to energy consumption considering weather?
  - Do you have any **recommendations of speed** to reduce power of one trip considering weather?
- **Cleaning:** Do you see based on the performance that the **hull and/or propeller** should be cleaned?
  - Additional question: Do you think it is worth cleaning?
- **Prediction**: Can you find a robust **prediction model** of shaft power (power used for propulsion)?
  - Input route optimization.



Additional sources

- Introduction Real vessel data challenge: http://data.hullpic.info/HullPIC2025_Mulheim.pdf
- Results from challenge: http://data.hullpic.info/HullPIC2026_Pontignano.pdf

Teknologi for et bedre samfunn

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